

Project Description for Portfolio

Project Title: Tameri Network

Overview: Tameri Network is a comprehensive, full-stack social platform built to seamlessly connect donors and visitors with orphanages. Designed with a scalable microservices architecture, the application facilitates community engagement through a robust social feed, secure real-time messaging, and a transparent donation system. It provides distinct user experiences for standard visitors and orphanage administrators, allowing for dynamic profile verification, needs publishing, and localized search.

Key Features:

- **Social & Community Feed:** A full-featured social feed supporting posts, comments, likes, reposts, and multimedia sharing.
- **Real-Time Messaging:** Instant peer-to-peer chat with real-time status updates, typing indicators, and message persistence.
- **Donation & Wallet System:** An integrated digital wallet allowing users to purchase "Stars" and send specific physical gifts (e.g., Clean Water, Groceries, Educational Supplies) to orphanages, integrating with Stripe and Orange Money.
- **Geolocation & Search:** Map-based search functionalities allowing users to find nearby orphanages and filter them by their specific published needs.
- **Orphanage Management:** Dedicated tools for orphanages to undergo verification, set up profiles, and track received donations and support metrics.
- **Push Notifications:** A reliable notification system to keep users engaged with recent activity, messages, and donation statuses.

Technologies Used

Frontend (Mobile App)

Built with a "Feature-First / Clean Architecture" approach to ensure maintainability and separation of concerns.

- **Framework:** Flutter & Dart (Cross-platform iOS & Android development)
- **Networking & API:** Dio (HTTP Client with custom interceptors for token refresh)
- **Real-Time Communication:** WebSockets (`web_socket_channel`)
- **Local Storage:** Flutter Secure Storage (for JWT tokens) & Shared Preferences
- **Mapping & Location:** `flutter_map`, `latlong2`, and `geolocator`
- **Media Handling:** `image_picker`, `video_player` / `chewie`, `audioplayers`, `cached_network_image`
- **Push Notifications:** Firebase Cloud Messaging (FCM)

Backend (Microservices Architecture)

Designed as a highly decoupled, event-driven microservices ecosystem using Docker.

- **Core Microservices Framework:** Python 3.10 with **FastAPI** (running on Uvicorn) for blazing-fast API performance.

- **Service Discovery:** Java 17 & **Spring Boot** (Eureka Server) for dynamic routing and service registry.
- **Primary Database:** **PostgreSQL 15** (Relational data storage for profiles, posts, donations, etc.).
- **Event Streaming / Message Broker:** **Apache Kafka** & Zookeeper (Used for asynchronous, inter-service communication and real-time events).
- **Object Storage:** **MinIO** (S3-compatible local storage for user media, avatars, and attachments).
- **Containerization & Orchestration:** **Docker** & **Docker Compose** (Defines and runs multi-container environments).
- **Services Implemented:** API Gateway, Profile Service, Post Service, Message Service, Donation Service, Notification Service, Search Service, Verification Service, Media Service, Admin Service, and Websocket Service.